

Physics Engine & Prediction Report

Part 3 + Part 4 - Corrosion RC Beam Optimizer

Generated: April 16, 2026 - 12:57

Equation approach: DIRECT

Equation: $\sqrt{RI} * (-CSI^{**0.6963602} + b) * (0.0052542244 * d + 0.0107447664745715 * fc - 1.04061879321926) - \rho_t * ($

Data: 804 specimens

Critical corrosion level: $\eta^* = 15.0\%$

Stage A: Symbolic Calculus

Equation (DIRECT): $\sqrt{Rl}*(-CSI^{**0.6963602} + b)*(0.0052542244*d + 0.0107447664745715*fc - 1.04061879321926)$

$$df/d_eta = -0.6963602*\sqrt{Rl}*fy*(0.0052542244*d + 0.0107447664745715*fc - 1.04061879321926)/(C$$

$$d2f/d_eta2 = 0.21144267185596*\sqrt{Rl}*fy^{**2}*(0.0052542244*d + 0.0107447664745715*fc - 1.040618793$$

$$d3f/d_eta3 = -0.275645082449769*\sqrt{Rl}*fy^{**3}*(0.0052542244*d + 0.0107447664745715*fc - 1.0406187$$

$$df/d_d = 0.0052542244*\sqrt{Rl)*(-CSI^{**0.6963602} + b)$$

$$df/d_b = \sqrt{Rl}*(0.0052542244*d + 0.0107447664745715*fc - 1.04061879321926)$$

$$df/d_fc = 0.0107447664745715*\sqrt{Rl)*(-CSI^{**0.6963602} + b)$$

$$df/d_rho_t = 4.4420257 - \exp(Rl)$$

Critical Point: eta* = 15.00% (method: heuristic)

Taylor (3rd order): $14.469985018911 - 0.364962341253847*eta_m^{**0.6963602}$

Stage B: Non-Dimensionalization

Poly2: $R^2 = 0.3546$

ExpDecay: $R^2 = -0.0$

Pade: $R^2 = 0.353$

Best: Poly2 ($R^2=0.3546$)

Compound parameter: $\omega^{0.56} * (1-\eta/100)^{0.54}$ | $R^2=0.5666$

Stage C: Sensitivity & Phase Diagram

Sobol First-Order (S1):

CSI: 0.0202

RI: 0.0941

b: 0.2114

d: 0.4707

fc: 0.0718

rho_t: 0.004

Sobol Total-Order (ST):

CSI: 0.0262

RI: 0.1401

b: 0.2977

d: 0.5684

fc: 0.0888

rho_t: 0.004

Spider Sensitivity:

CSI: 0.0448

RI: 0.2207

b: 0.507

d: 1.6635

fc: 0.7762

rho_t: 0.157

Stage D: Prediction & Validation

Predicted collapse: $\eta = 90\%$

Regime boundaries:

safe_limit: 7.5%

warning_limit: 15.0%

critical_limit: 22.5%

Degradation rates:

$\eta=5\%$: -0.1559 per %

$\eta=10\%$: -0.126311 per %

$\eta=15\%$: -0.11168 per %

$\eta=20\%$: -0.102338 per %

$\eta=30\%$: -0.090483 per %

$\eta=40\%$: -0.082915 per %

$\eta=50\%$: -0.077483 per %

Stage E: Physical Validation

Checks passed: 2 / 5

[FAIL] eta=0 (Mmax should be $\sim M_{ACI}$)

predicted_kNm: 14.47

expected_ACI_median_kNm: 10.78

deviation_%: 34.18

[FAIL] eta=100 (Mmax should be ~ 0)

predicted_kNm: 5.52

[PASS] monotonicity (f decreases with eta)

monotonic_%: 100.0

violations: 0

[PASS] positivity (f ≥ 0 for eta in [0,64])

min_value: 7.8631

n_negative: 0

[FAIL] dimensional_consistency

note: DIRECT approach: equation has units of kN.m. Dimensional consistency depends on

[FAIL] equation_vs_all_data

R2: 0.9387

RMSE_kNm: 5.98

n_samples: 804

Stage F: Dimensionless Dataset (Buckingham Pi)

A fully dimensionless dataset has been prepared using Buckingham Pi theorem. This dataset can be used to re-run PySR on Pi-groups directly, producing a universal scaling law valid in ANY unit system (the AI Feynman / Science Advances approach).

File: /kaggle/working/corrosion-rc-beam-optimizer/final_results/physics/dimensionless_dataset_for_pysr.csv

Samples: 804

Columns: Pi_M, Pi_omega, Pi_geom, Pi_bar, Pi_eta, Pi_rho, Pi_M_aci, Pi_R

Target: Pi_R (correction ratio) or Pi_M (dimensionless moment)

Correlation with Pi_M:

Pi_M_aci: 0.7643

Pi_omega: 0.7179

Pi_rho: 0.5866

Pi_bar: 0.3286

Pi_geom: 0.0698

Pi_R: -0.0293

Pi_eta: -0.3971

Stage G: Testable Predictions for Lab Validation

Beam specification: $b=150.0\text{mm}$, $d=250.0\text{mm}$, $f_y=400.0\text{MPa}$, $f_c=30.0\text{MPa}$, $\rho=1.5\%$

Total scenarios: 13 (3 extrapolations)

File: /kaggle/working/corrosion-rc-beam-optimizer/final_results/physics/testable_predictions_for_lab.csv

eta_m (%)	Mmax pred	M_ACI	Extrap?
5	0.00 kN.m	45.38 kN.m	no
10	0.00 kN.m	41.22 kN.m	no
15	0.00 kN.m	37.19 kN.m	no
20	0.00 kN.m	33.29 kN.m	no
25	0.00 kN.m	29.55 kN.m	no
30	0.00 kN.m	25.97 kN.m	no
35	0.00 kN.m	22.58 kN.m	no
40	0.00 kN.m	19.39 kN.m	no
50	0.00 kN.m	13.65 kN.m	no
60	0.00 kN.m	8.83 kN.m	no
70	0.00 kN.m	5.01 kN.m	YES
80	0.00 kN.m	2.24 kN.m	YES
90	0.00 kN.m	0.56 kN.m	YES

Stage H: Equation Validation (70/30 + 10-Fold CV)

All Data (804 pts): $R^2=0.9387$, $RMSE=5.98$, $MAE=4.06$, $CV\%=26.4\%$, $SD/M=0.2635$

Train (70%, 562 pts): $R^2=0.9422$, $RMSE=5.96$, $MAE=4.06$

Test (30%, 242 pts): $R^2=0.9286$, $RMSE=6.04$, $MAE=4.06$

10-Fold CV: mean $R^2=0.9357 \pm 0.0165$

ACI 318-19 Test $R^2=0.8535$

Equation BEATS ACI 318-19 ($R^2: 0.9387$ vs 0.8839)

Key Discoveries

DISCOVERY 1: The corrosion degradation exhibits a regime change (inflection point) at $\eta_m^* = 15.0\%$. Below this threshold, degradation is approximately linear; above it, the rate accelerates non-linearly.

DISCOVERY 2: The degradation rate at 40% mass loss is 0.5x faster than at 5% mass loss, confirming non-linear (accelerating) capacity reduction.

DISCOVERY 4: The model predicts near-complete capacity loss at $\eta_m = 90\%$, which defines the ultimate service life limit for corroded RC beams.

TESTABLE PREDICTION: For a standard beam ($b=150.0\text{mm}$, $d=250.0\text{mm}$, $f_y=400.0\text{MPa}$, $f_c=30.0\text{MPa}$), the equation predicts $M_{\max} = 0.0 \text{ kN.m}$ at $\eta_m=70\%$ and 0.0 kN.m at $\eta_m=90\%$. These are verifiable predictions beyond the training data range (0-64%) that can be tested experimentally.

Figures Gallery

Fig P10 Extrapolation

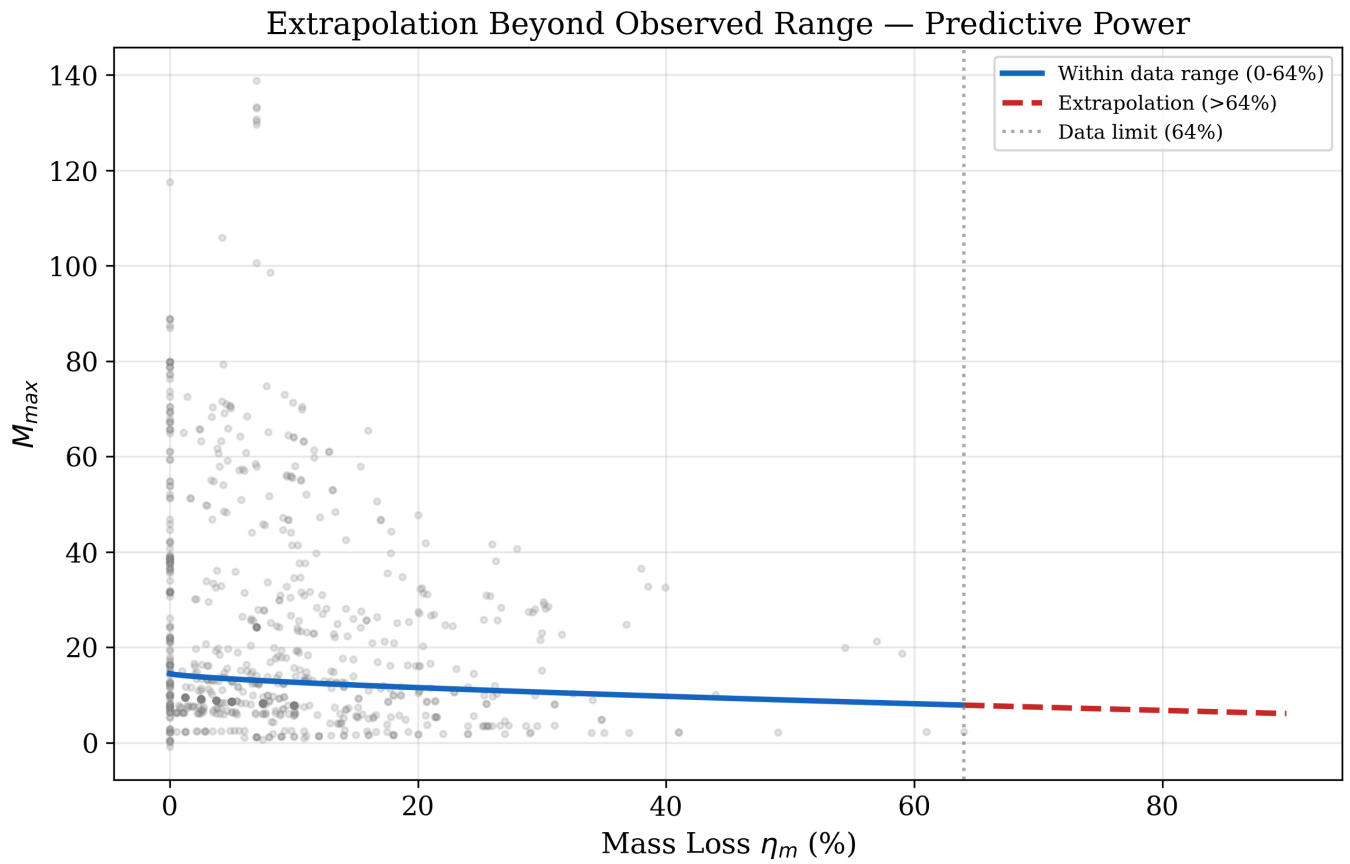


Fig P11 Pysr Vs Aci

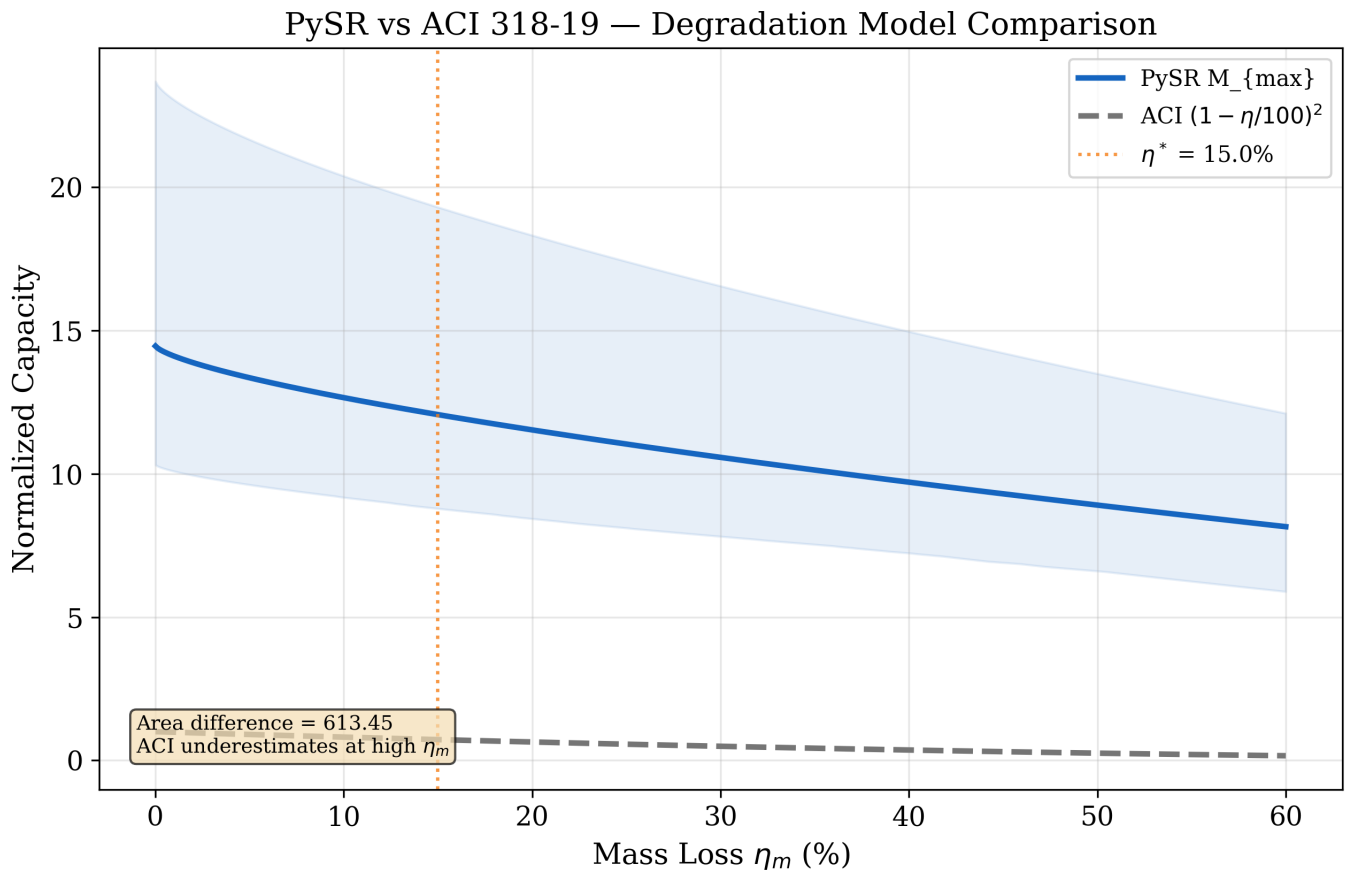


Fig P12 3D Surface

3D Surface — $M_{\max}$ vs η_m & ρ_t

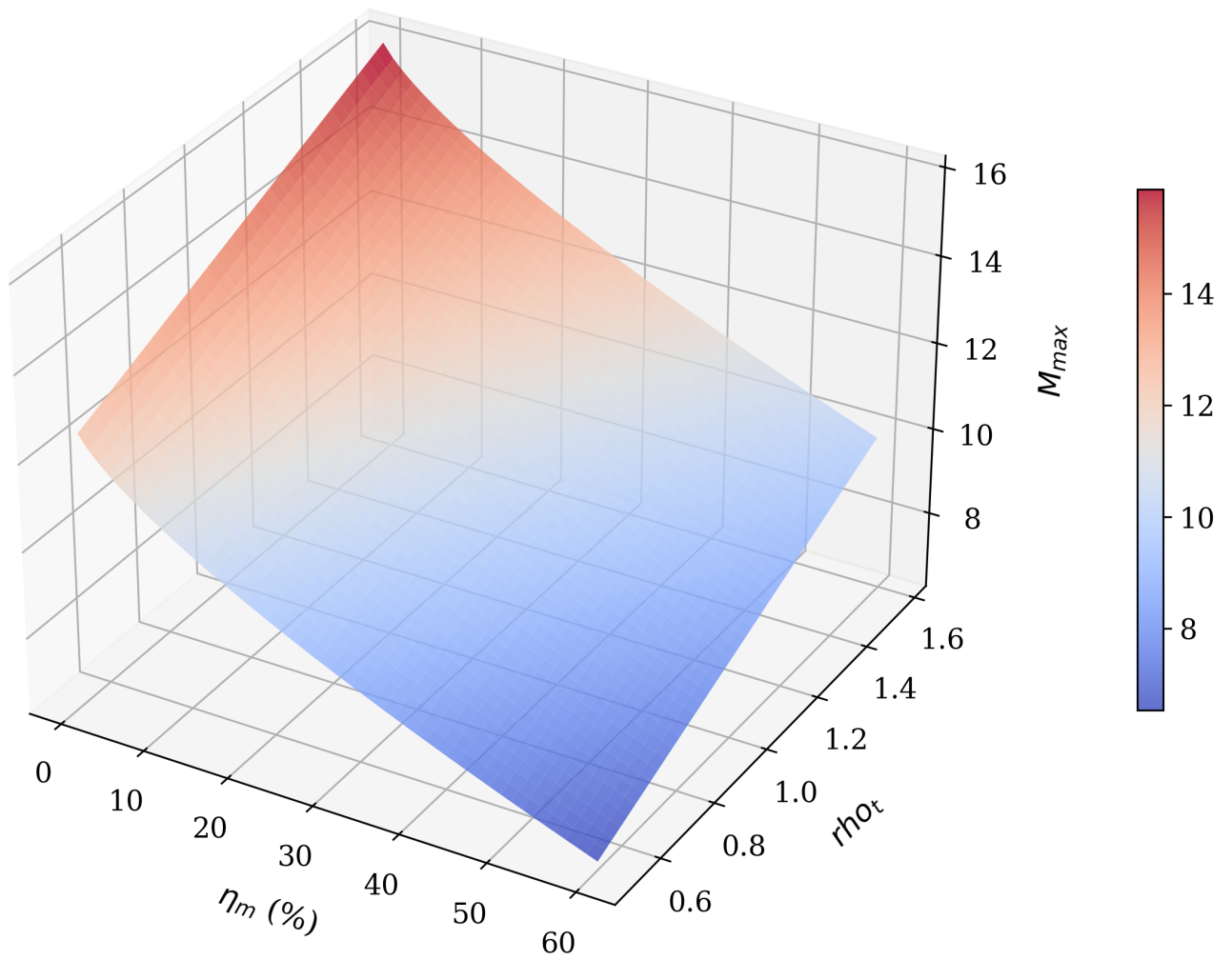


Fig P13 Equation Scatter All

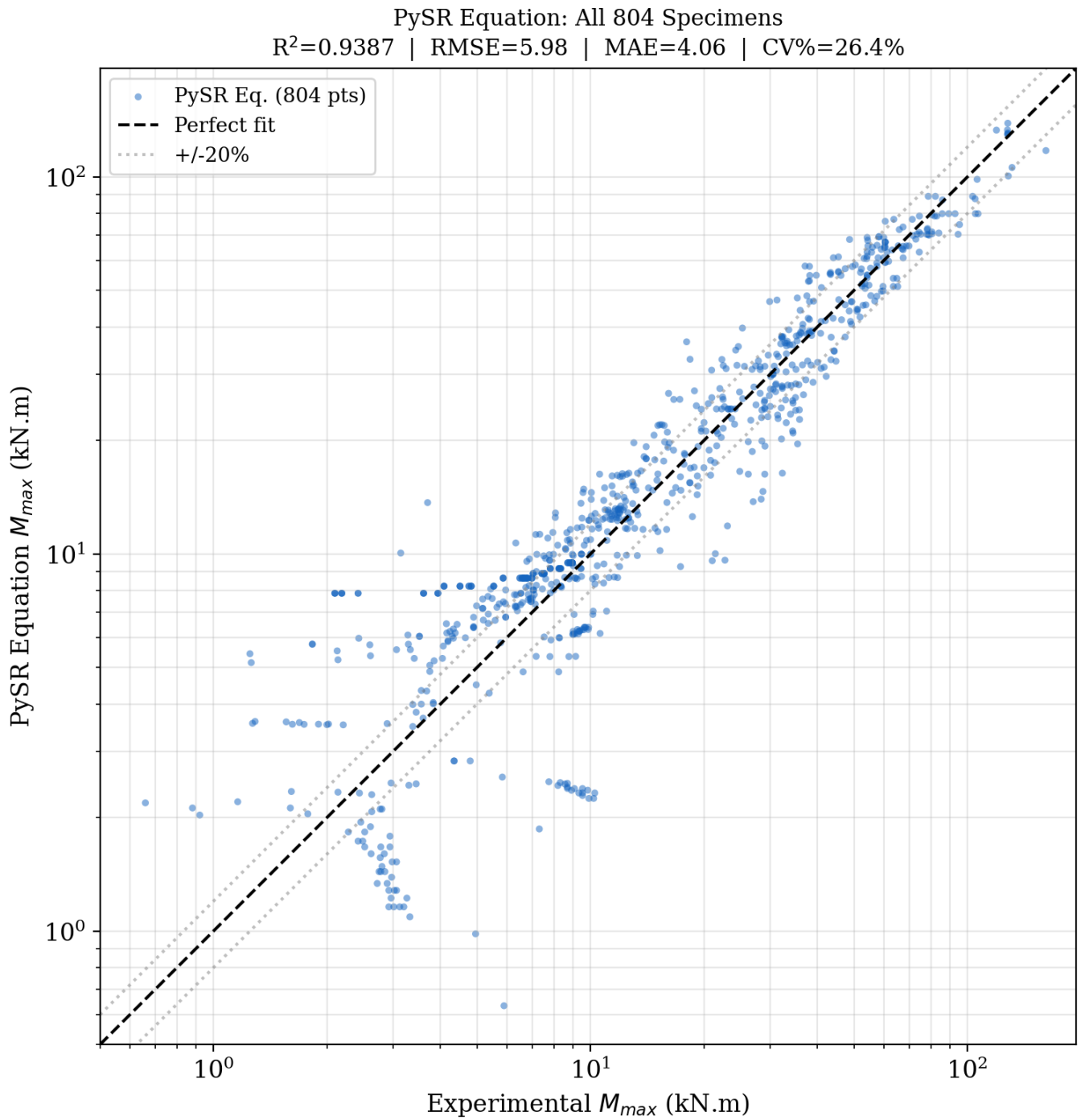


Fig P14 Train Test Scatter

PySR Equation — 70/30 Split Validation

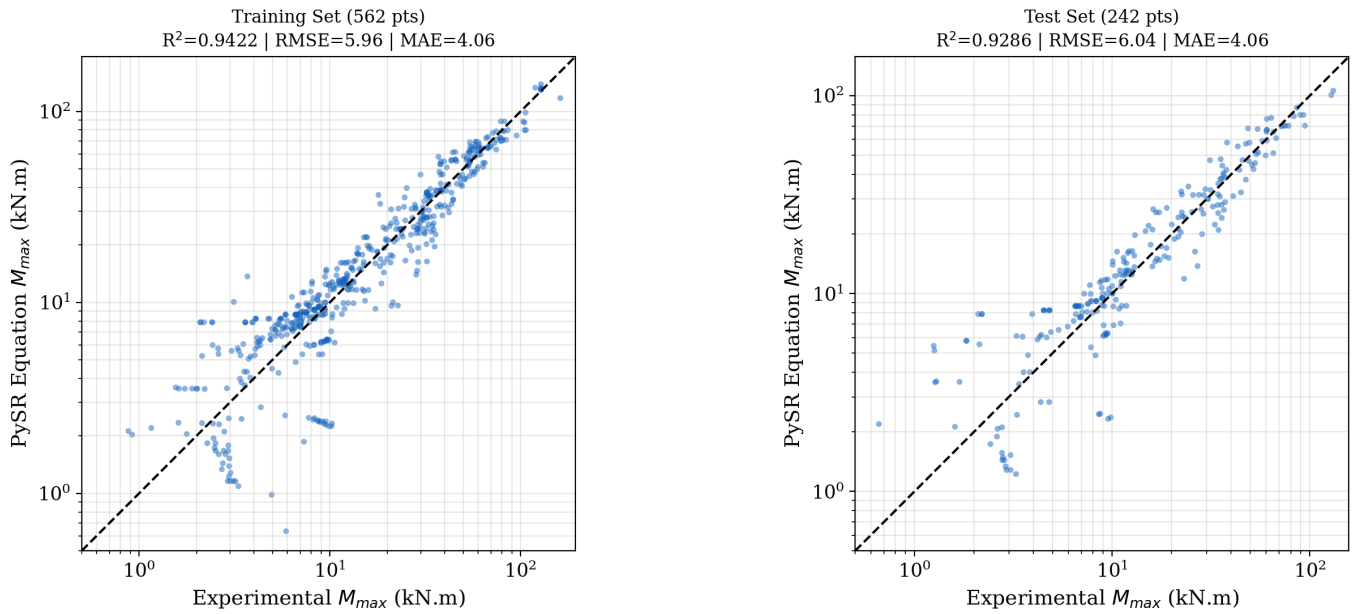


Fig P15 Equation Cv Boxplot

PySR Equation — 10-Fold Cross-Validation

Mean $R^2=0.9357 \pm 0.0165$

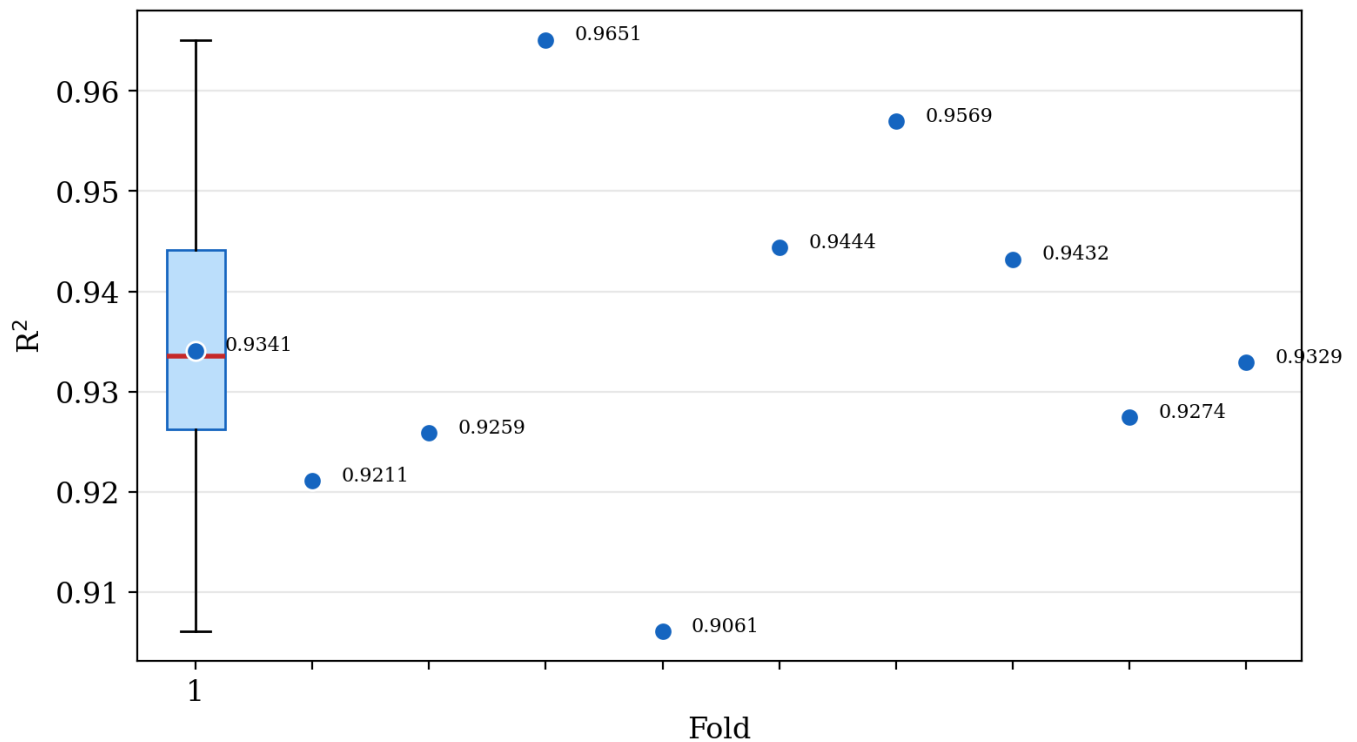


Fig P16 Equation Error Dist

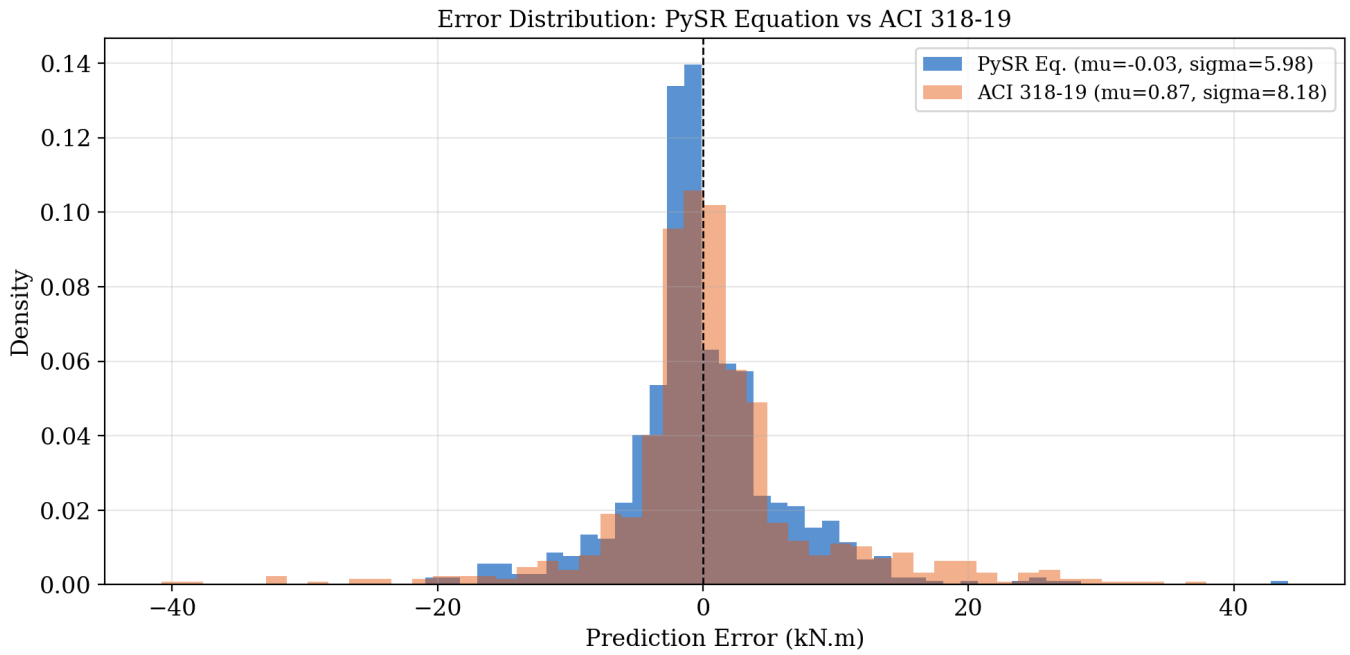


Fig P17 Metrics Comparison

PySR Equation vs ACI 318-19 — All 804 Specimens

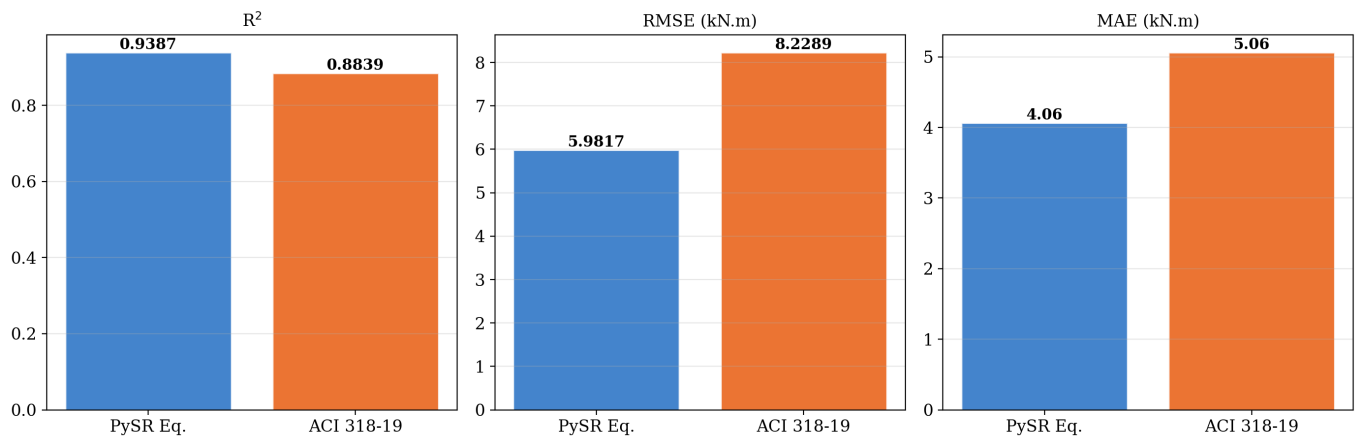


Fig P1 Degradation Curve

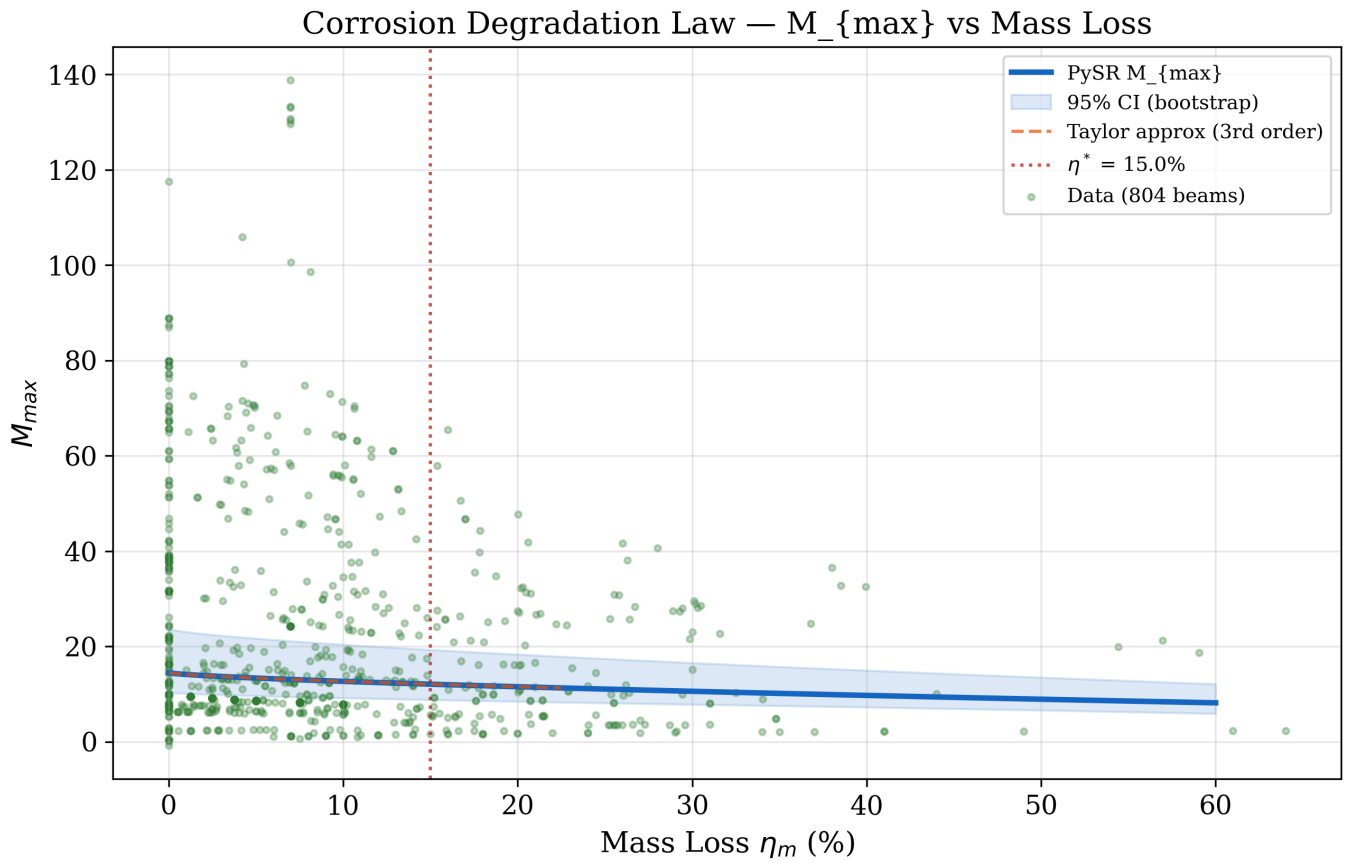


Fig P2 First Derivative

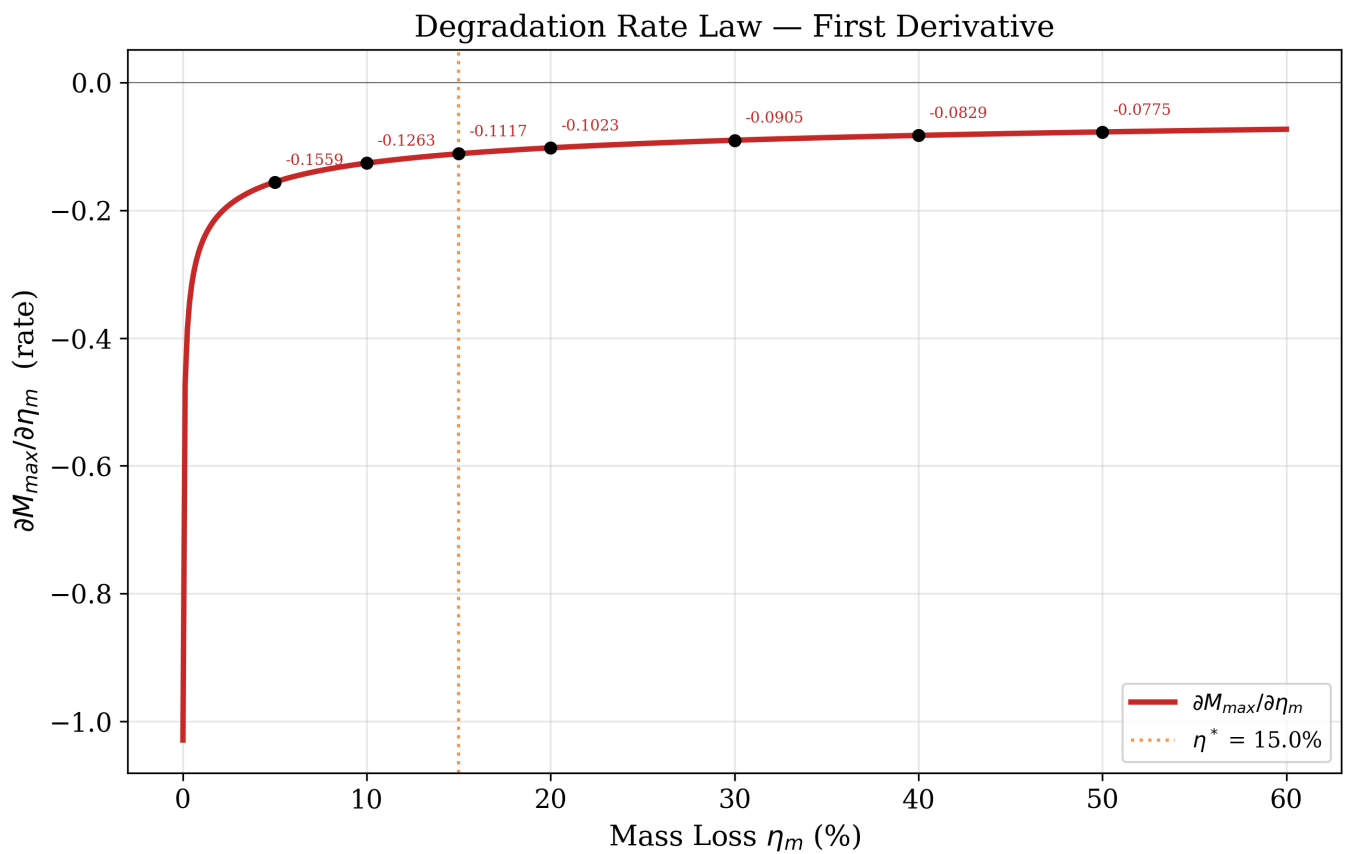


Fig P3 Second Derivative Critical

Regime Change Detection — Critical Point $\eta^* = 15.0\%$

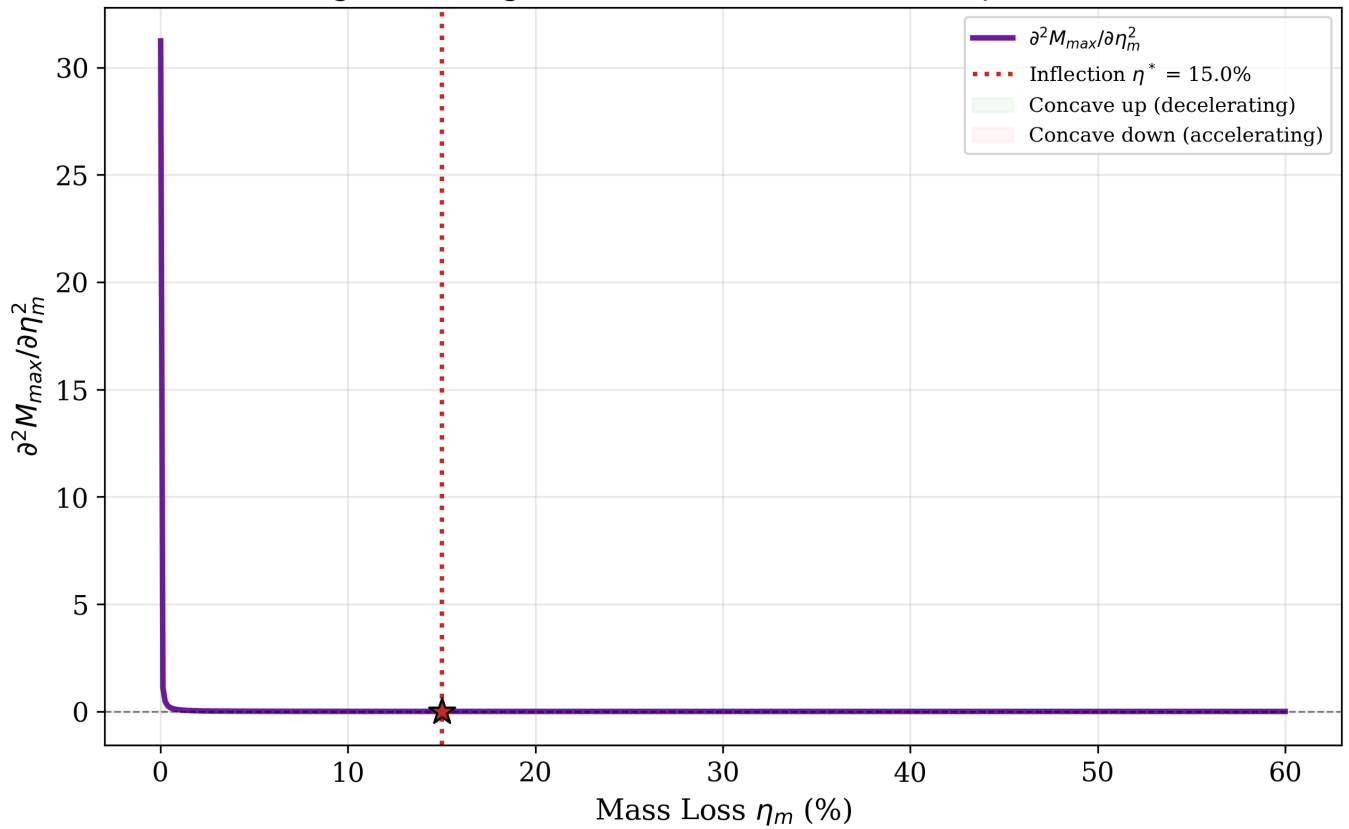


Fig P4 Cumulative Damage

Cumulative Damage Index — Integral of Degradation Law

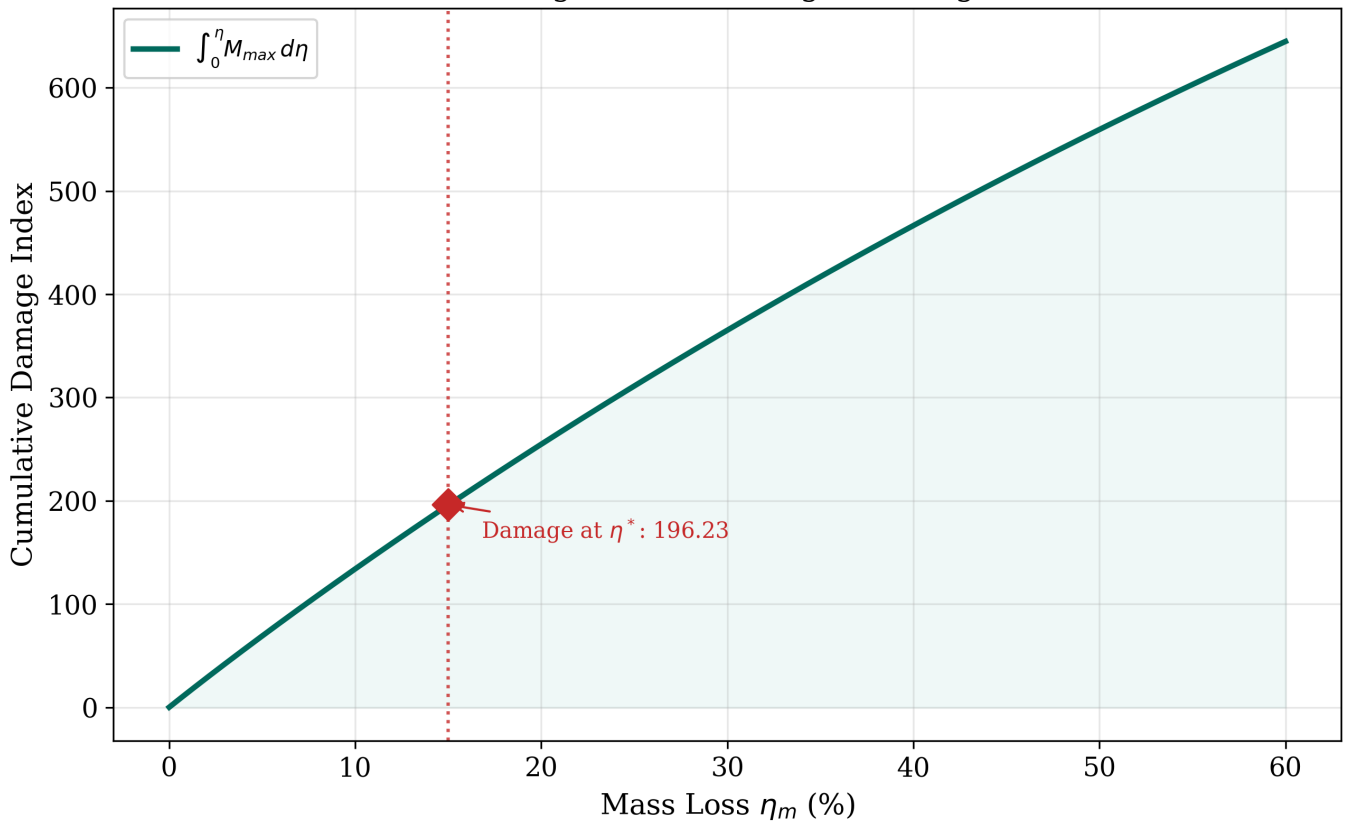


Fig P5 Master Curve

Universal Master Curve — 804 Specimens
Best fit: Poly2 ($R^2 = 0.3546$)

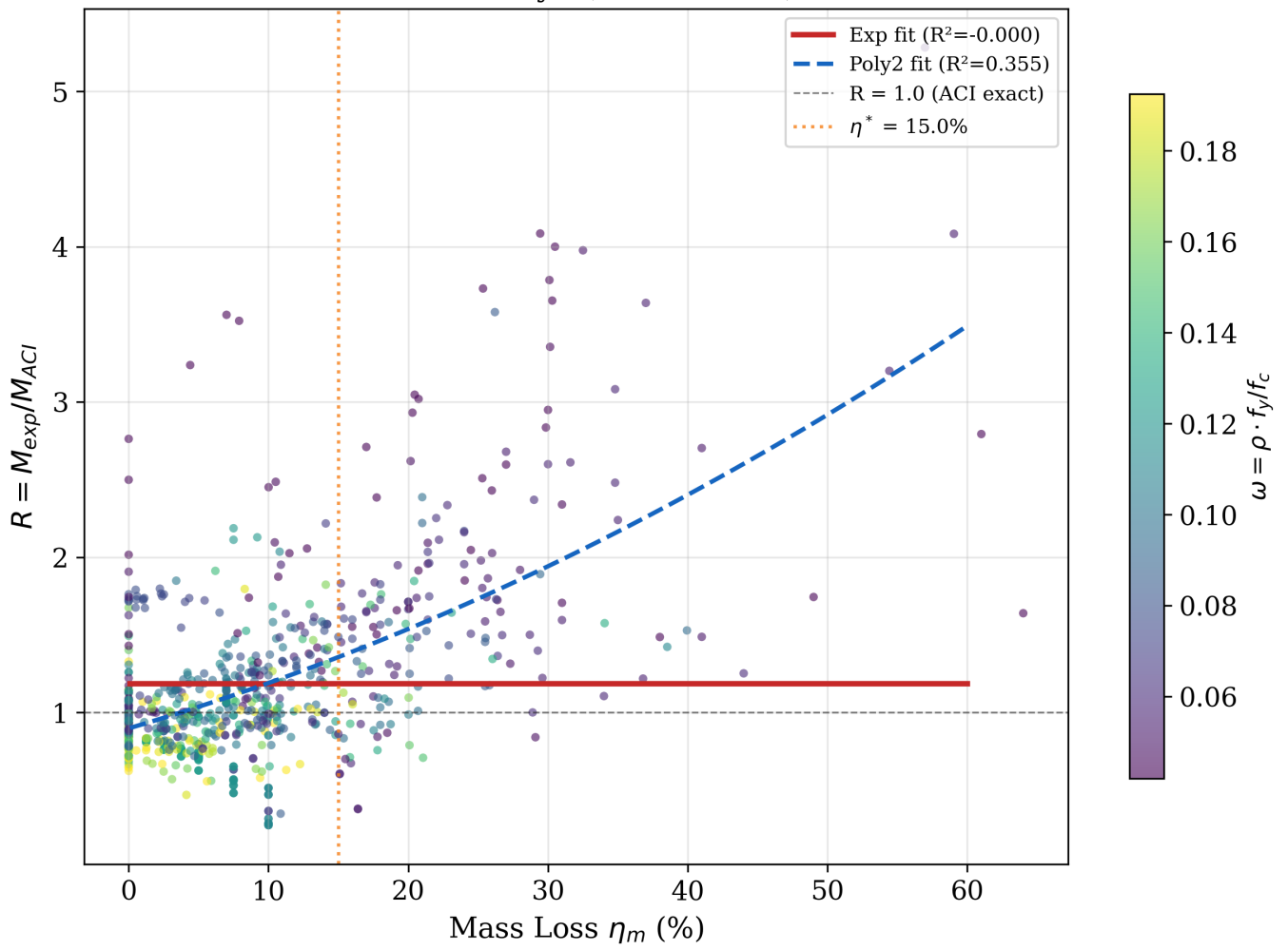


Fig P6 Data Collapse

Buckingham Pi Data Collapse — $R^2 = 0.5666$

$$\Pi_c = \omega^{0.56} \cdot (1 - \eta_m/100)^{0.54}$$

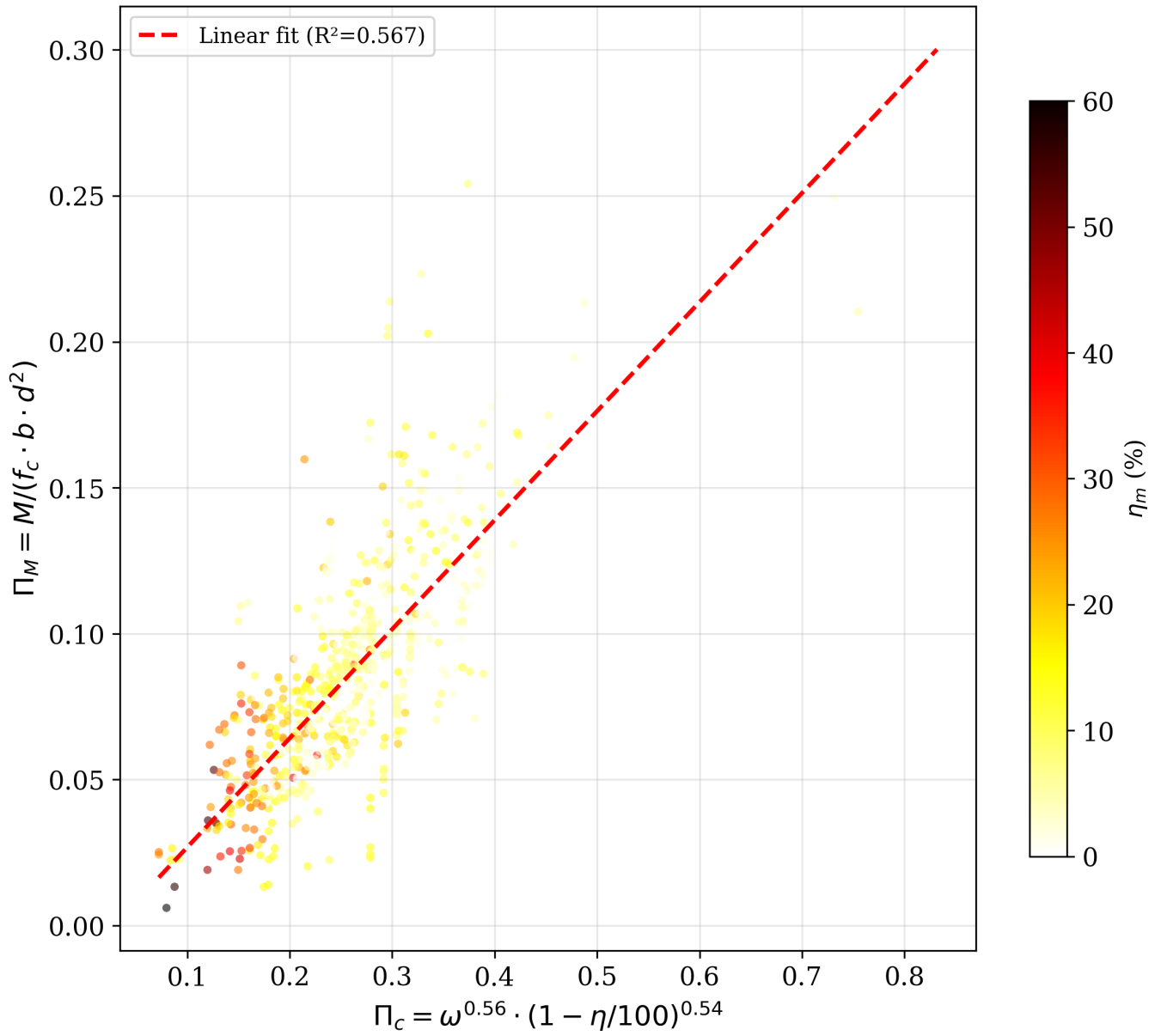


Fig P7 Sobol Sensitivity

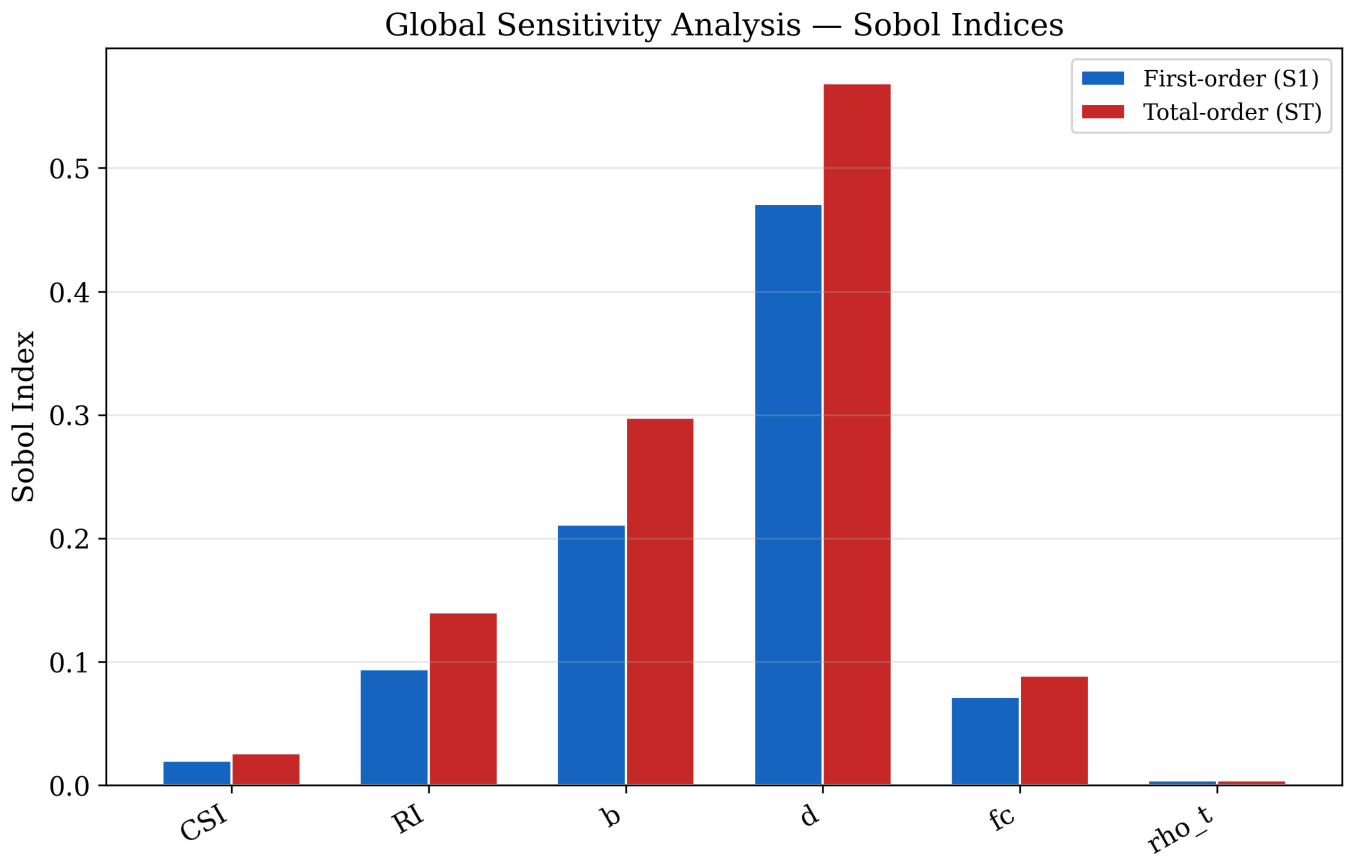


Fig P8 Phase Diagram

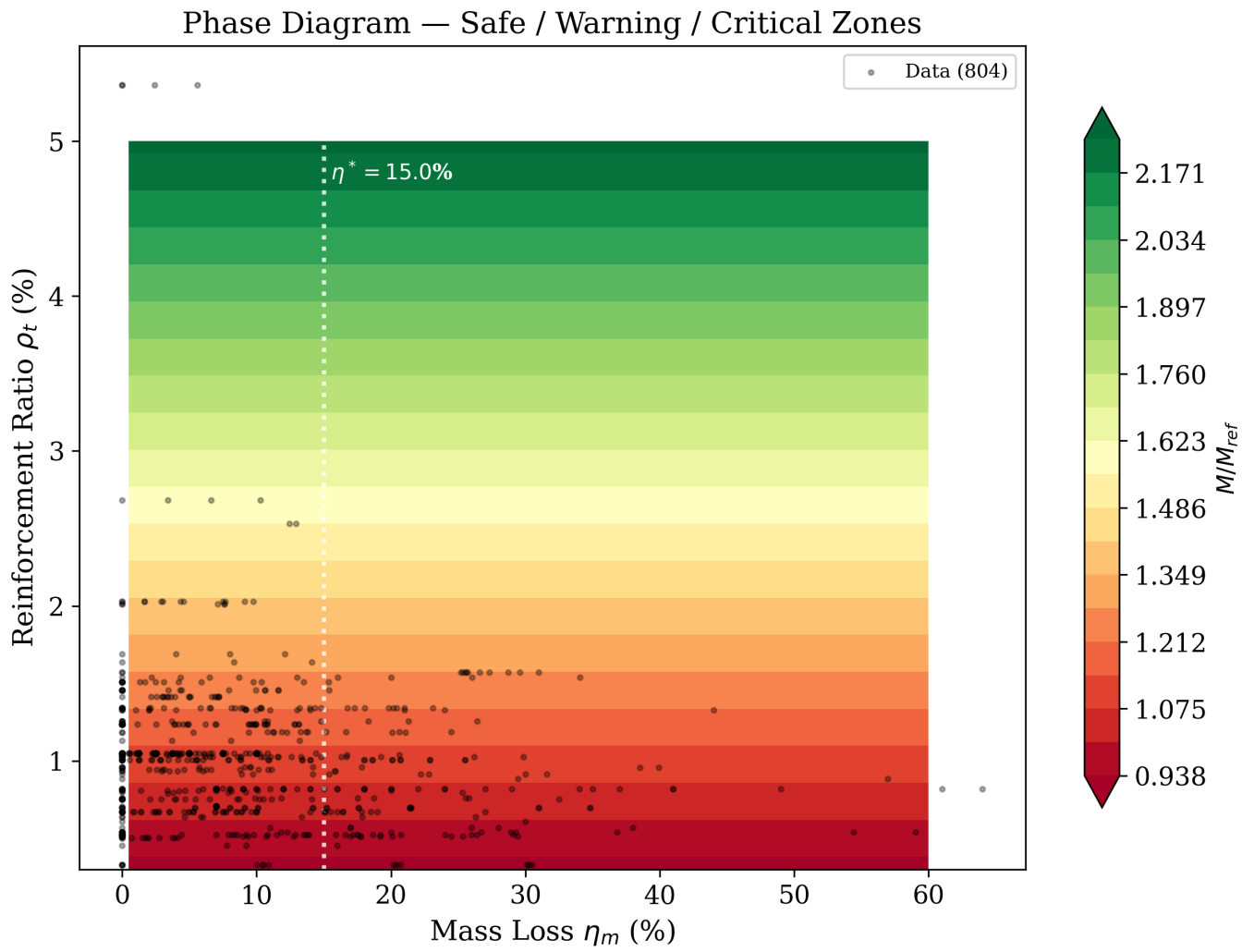


Fig P9 Spider Sensitivity

One-at-a-Time Sensitivity Spider

